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Genesis of coexisting itinerant and localized electrons in Iron Pnictides

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Abstract We show how the general features of the electronic structure of the Fe-based high-Tc superconductors are a natural setting for a selective localization of the conduction electrons to arise. Slave-spin and Dynamical mean-field calculations support this picture and allow for a comparison of the magnetic properties with experiments.

Keywords Iron Pnictides · Orbital-selective Mott transition

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1 Introduction

Since 1986, each time a new "high-temperature" superconductor has been discovered, it has been a natural temptation to compare the new materials to the cuprates, looking for unifying aspects which could shed light on the key mechanism for high-temperature superconductivity. This temptation is probably stronger then ever in the case of the iron-based "pnictides", the second family of transition-metal-based superconductors with critical temperature exceeding 50K[1].

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The two classes of materials share indeed several features among which we may quote the structure, which is quite similar, with FeAs layers playing a similar role of CuO planes, and the observation that in both cases superconductivity appears doping a magnetically ordered parent compound, and follows a dome behavior as a function of doping.

An important different lies in the nature of the parent compound. In the cuprates the half-filled system is an insulator which is usually considered a Mott insulator, testifying the central role of electron correlations (Even if it has been recently proposed that the degree of correlation can be smaller than usually believed[2]). The undoped pnictides are instead still metallic, despite the magnetic ordering, suggesting a weaker degree of electronic correlations. The actual strength of correlation in pnictides is lively debated: while there is no obvious reason to invoke correlation as the key player in the game as it happens in cuprates, it has been proposed that these compounds are actually close to a Mott transition[3].

Another peculiarity of pnictides is that, according to density-functional theory calculations, all the five d-bands of iron are relevant as opposed to the single copper band of the cuprates. Multiple bands crossing the Fermi level are confirmed by angular resolved photoemission. The relevance of multiband effects can solve the discrepancy between different estimates of the role of correlations. Indeed electron correlation can induce a much richer physics in multiband systems with respect to single band. In particular, different bands can be affected differently by correlations, and eventually one can have an orbital-selective Mott transition (OSMT), in which some bands become insulating and others are metallic. Signatures of an orbital-selective behavior have been detected by angle-resolved